AGILE

S/w or sprint design

HLD (overall architecure of entire sprint s/w)

Multiple LLds (indepth logic of each module in sprint s/w)

Eg:-

Login → Bills pay → logout

Login ↓ Money transfer ↓ logout

Eg:-

user ↓ userid & password ↓ login → DB

login → Invalid → Error messag → login

login ↓ valid ↓ nextscreen

Agile principles:-

1) Individuals & Interactions over process and tools.
2) Working software over comprehensive documentation.
3) Customer collabaration over contract negotiation.
4) Responding to change over following a plan.

Agile model

S/w bidding

↓

kick of meeting

Product owner

Product backlog grooming mtng

Refinement mtng

sprint retrospective

sprint release

↓

support

maintenance

Sprint planning mtng

Dev/testing

Daily scrum

Sprint review mtng

components in Agile process:—

1) Agile Ceremonies

↳ Kick of meeting

→ Product backlog grooming mtng

→ Sprint planning mtng

→ Daily/standup meeting/scrum mtng

→ Sprint review meeting.

→ Sprint retrospective mtng

→ Refinement mtng

→

2) Artifacts

→ product Backlog

→ sprint Backlog

→ HLD

→ LLD → program - Testcases, Test summary report - Automation script.

→ RTM (Requirement traceability matrix)

→ Burndown chart.

3) Roles:-

CEO → takes care about all the proposals of projects from client

client

stakeholders → client representatives (or) the team members

Product owner

scrum master

Dev

QA

Ceremony in agile Method	Roles to be involved	purpose
1) Kick of meeting.	CEO, customer stakeholders	→ To select product owner → No time limit for meeting.
2) Product backlog grooming mtng → req gathered from client	Product owner, Stakeholders	→ To gather all req as user stories. → If any userstory is epic, divide that user story into multiple user stories. → PO can take help of subject matter expert if required. → Finalize Product backlog document. → No time bound.
3) Sprint planning meeting	product owner, Stakeholders, Scrum master, Scrum Team	→ selects some userstories from all user stories depends on size, complexity and priority, → Finialize sprint backlog. → No time limit. sprint backlog → selective req to work in ~~sprint~~
4) Daily standup / Daily scrum mtng	PO, SHS, SM, ST	→ To discuss about what we have done yesterday, what we are going to do today, and if any impediments. → 15-20 min.
5) Sprint review meeting	customer, PO, SHS, SM, ST	→ To collect feedback from customer on finished project. → Follow alpha / Beta / gamma manner to collect feedback.

		alpha means providing sprint demo to customer by operating sprint from our people. → beta means giving chance to customer to operate sprint in their environment. → gamma means providing demo to customer and giving chance to customer to operate sprint → no time bound.
6) sprint retrospective mtng	Scrum master, scrum team & product owner	→ To know challenges related to scrum team members before going to next sprint → 1-2 hr is time box.
7) product backlog refinement mtng	customer, PO, SHs, SM, ST	→ To know the changes to be performed in user stories of product backlog. → 1-2 hrs is time box.

we will go through the user stories in the next sprint whether any changes is there to implement in userstories

Artifacts:—

Artifacts :-

Roles :—

1) CEO :— The ceo is responsible to receive the proposal of a project from the client.

→ He is also responsible to appoint. project manager for the project.

2) Client / customer :-

→ The main responsible of client to prepare & Project proposal. He is also responsible to provide the requirement to the project team.

→ He is also responsible to involve in the sprint review meeting to provide an early feedback and also responsible to involve the stakeholders for the project.

3) Stake holders :-

→ Stakeholders are considered as representative of customers.

→ He will involve in the product backlog grooming meeting to the provide the req along with SME [subject matter experts]

→ He also involve in the sprint review meetings to provide the feedback.

→ He also involve in the sprint retrospective meetings to discuss about the challenges faced by the team and what well in the team.

→ He also involve refinements meeting to go through the feature sprint user stories to make sure all the requirements are proper.

Product owner:-

→ Product owner is responsible to gather the requirements through product backlog grooming meeting from the client

→ He is responsible (for the) to prepare userstories based on the complexity and prioritization of userstories in the sprint.

→ He is responsible to involve in the daily standup meeting sprint retrospective meetings sprint refinement & Review meetings.

Scrum master:-

→ scrum master is responsible to manage the team

→ He involve in the daily standup mtng sprint review mtng, sprint retrospective mtngs.

→ He is responsible to clear the team level impediments (problems).

Scrum Team:-

→ scrumteam size is mostly 7 to 10 members. different roles like developers, database developer, QA people. together to form a scrumteam.

→ The dev team basically involves in analysis of req provided by the client and involve in the development of programs and integration of developers programs. They are also responsible to perform the unit testing, Integration testing & release the developed built to the 'QA Team'.

→ The main responsibility of the QA Team is to analyse the requirements provided by the either product owner or BA team or developer and involve in the preparation of test scenarios & testcases.

→ QA Team also perform testcase execution & bug raising & verification of the bugs and also involve in the automation scripting. automation executing preparation of summary reports & bug reports.

Artifacts:- (documents)

Product backlog:-

Product backlog is a place where we are going to store all the userstories related to an application.

→ In product backlog the userstories of prioritised based on the complexity & client decision.

Sprint backlog:-

Sprint is an duration of certain time that helps to complete requirements that we have takes into the sprint. mostly we have one week of sprint, 2 weeks of sprint, 3 weeks of sprints, 4 weeks of sprint.

→ In sprint we are going to take certain amount of userstories from product backlog based on the capacity of team.

HLD & LLD

→ High level design & low level design documents are diagrammatical representation of an application or features.

→ HLD represents overall architecture of the application which will be prepared by technical architects ~~or~~ leads in the teams.

LLD represents indepth design of a feature or sprint or userstories which are prepared by the leads or senior developers in the team.

Programs:-

Programs are set of the code return by the developers based on the LLD's that represents business requirements.

→ once the development of code is completed team will be perform the integration b/w different programs.

Test cases, Test scripts, Bug reports, Automation scripts, Test summary.

→ The QA team will write the test cases based on the requirements provided by the Client.

→ Test scenario is a one line description what we are going to execute.

→ Test steps test describes how we are going to execute scenario.

→ Team is responsible to perform the automation based on the scope defined using different languages like Java, C#, Python, Angular, php etc.,

→ Team is also responsible test summary report once the overall execution is done and also responsible to prepare bug reports.

Requirement Traceability matrix:-

→ It is an document which defines mapping b/w user stories against test cases via bugs

→ It helps to define the test coverages.

→ There are two different types:-

1) Forward requirement traceability matrix.

2) Backward " "

1) Forward requirement TM:-

Defines the mapping b/w userstories, ~~test cases~~, along with test case via bug.

2) Backward requirement TM:-

Defines mapping b/w via test cases with user stories.

Burn down chart:-

→ Burn down chart define that estimate work & work done by team.

→ Burndown chart is mainly handled by the project manager to view the sprint status.

→ It is an graphical representation based on the capacity estimated work and work done by the team.

Testing stages

Testing stages	Fish Model	V-Model	Agile scrum model
1) Document testing	Prepared by one team & reviewed by another team	Prepared & reviewed by the same team	Product owner can prepare & review documents, but approved by customer.
2) unit testing	coding by one team & unit testing by other team	coding & testing done by same team	coding & unit testing done by same team approved by stakeholders.
3) Integration testing	programs inter connection by one team & Integration testing by other team,	programs Interconnection & Integration testing by same team.	Program interconnection and integration testing by same team & approved by stakeholders.
4) Software testing (or) Sprint testing	S/w testing by seperate team	S/w testing by seperate team	S/w testing by testers in scrum team
5) Acceptance testing	Real/ model customer	Real/model customer	Real /model customer

Testing stages	Fish model	V-model	Agile/Scrum
Release testing	Release team	Release team	Release team
Support & Maintainance	Same team/CCB	same team/CCB	same team/CCB

Document testing

Document testing is mainly done by the product owner or BA.

→ In the product backlog we have all the user stories that are related to the entire application which are segregated based on the priority and

→ sprint backlog is a place where we are going to pull certain amount of userstories based on priority complexity, capacity of the team

→ The same sprint backlog will get reviewed by product owner.

→ Based on the sprint the technical architect (or) lead will prepare the HLD & LLD.

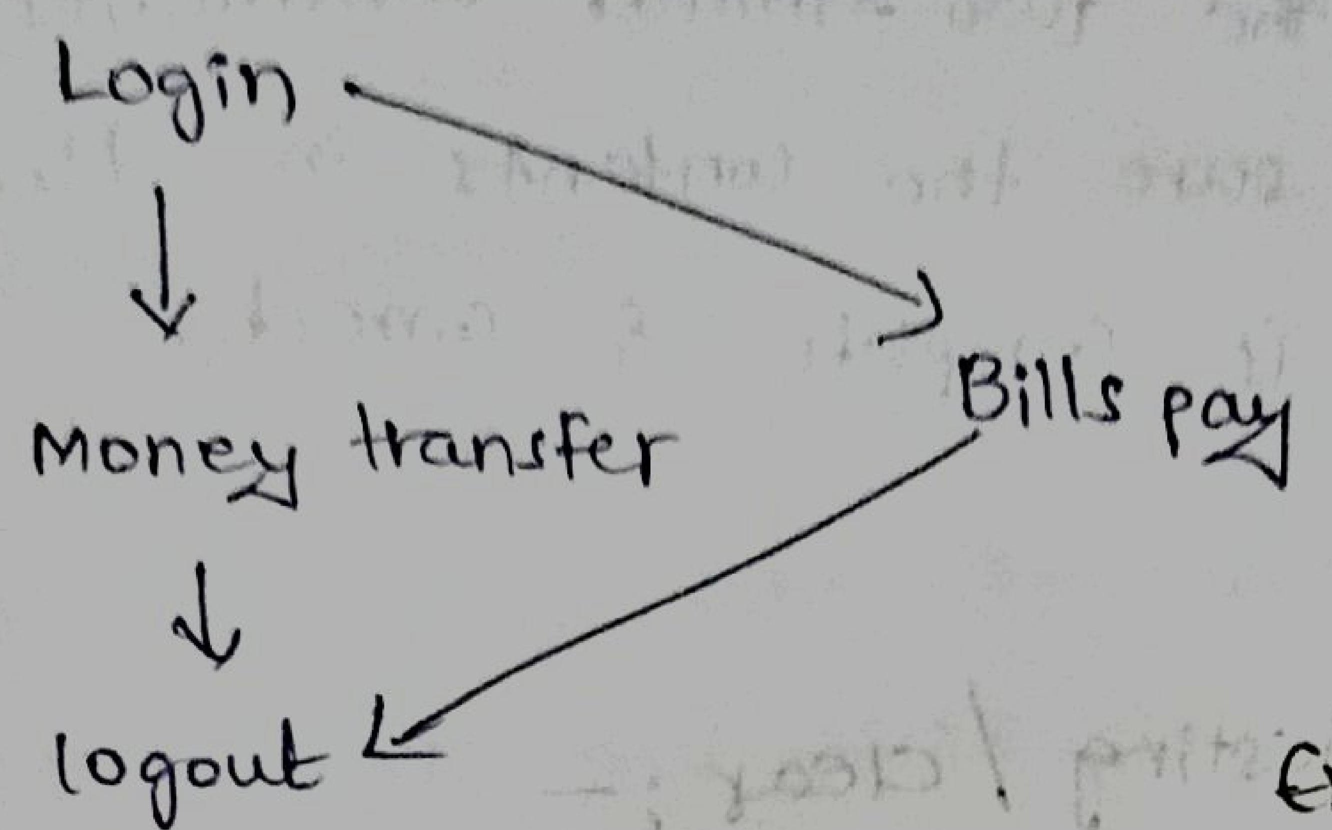
→ HLD defines the overall architecture of an entire sprint and LLD defines the indepth logic of each module (or) an userstory in the sprint.

S/w or sprint design

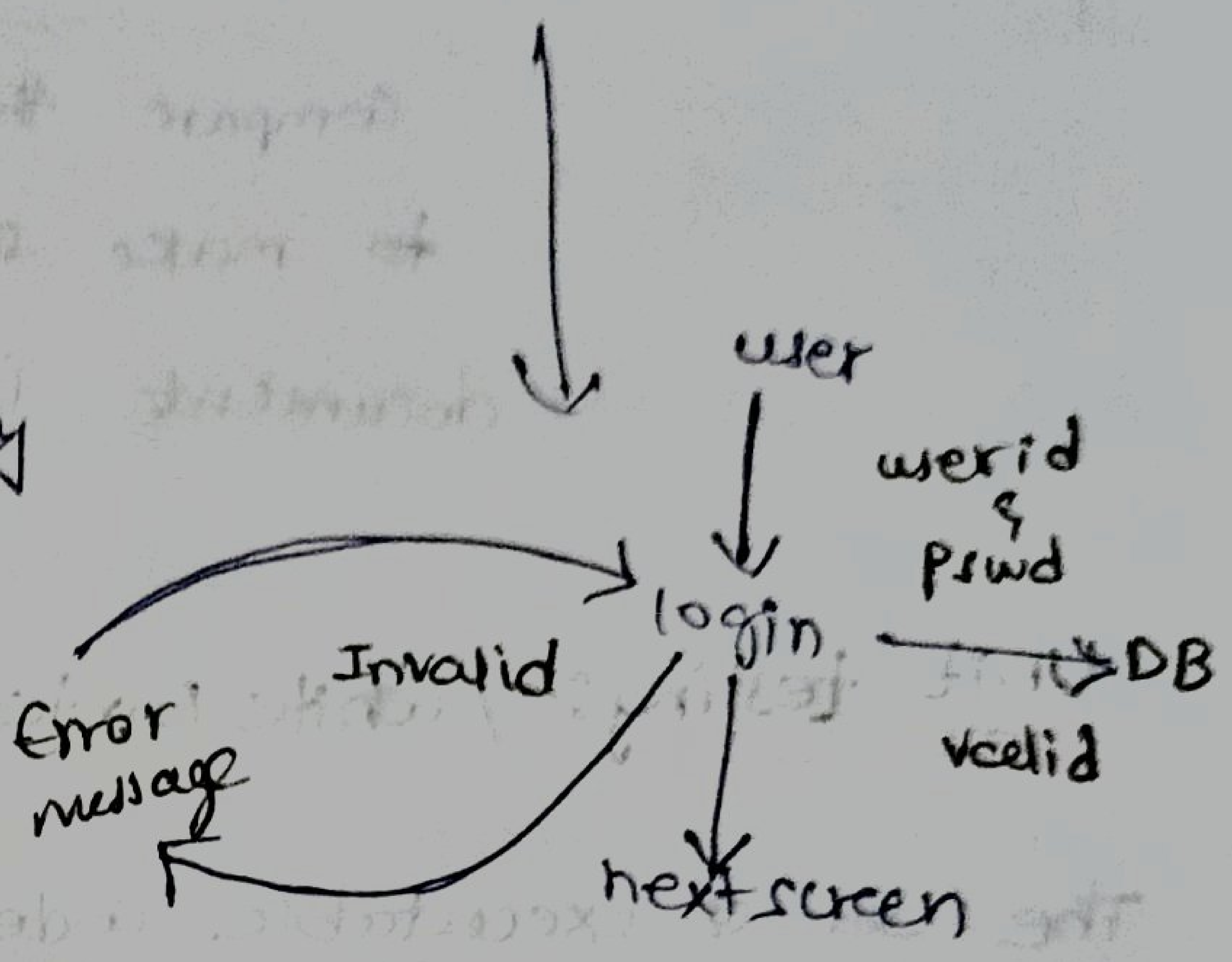
HLD (overall architecture of entire sprint s/w)

Multiple LLds indepth logic of each module in sprint s/w.

Ex:- HLD



Ex:- LLD of login



All the above mentioned document need to get reviewed by the respective people to make sure all the document are reviewed for its complete and correctness.

→ In order to review the team will follow the below review testing techniques or document testing techniques or static testing techniques.

1) Walkthrough:- In the walkthrough the team will review the entire line by line information in the requirements to make sure it is complete and correct.

→ complete in the sense without any misses and correct in the sense without any mistakes.

2) Inspection:- Here the team will concentrate on the main points to make sure the entire document is complete and correct.

3) Peer review:- In the Peer review the team will compare the two similar documents to make sure the contents in the document is complete & correct.

Unit testing:-/ white Box testing / clear:-

The set of executable code is called programs every block of code return by the developed will perform unit testing to check accuracy of the code whether it is working as per the expected result. In Order to perform unit testing the developers will follow certain techniques called white Box testing techniques or clear box testing techniques or open box testing techniques.

1) Basic path coverage:-

In these technique the developers will execute each and every block of the code to make sure the programs or running without any compile time errors.

2) Control Structure :-

In this technique the developers will execute each and every line of the code to make sure it is working as per the expected result. Here the developers will perform a process called debugging.

3) Program Technique coverage:-

In this technique the developers will execute every bit and piece of code to make sure the response is accurate and fast. mainly the developers will concentrate on speed of execution of the application (performance testing).

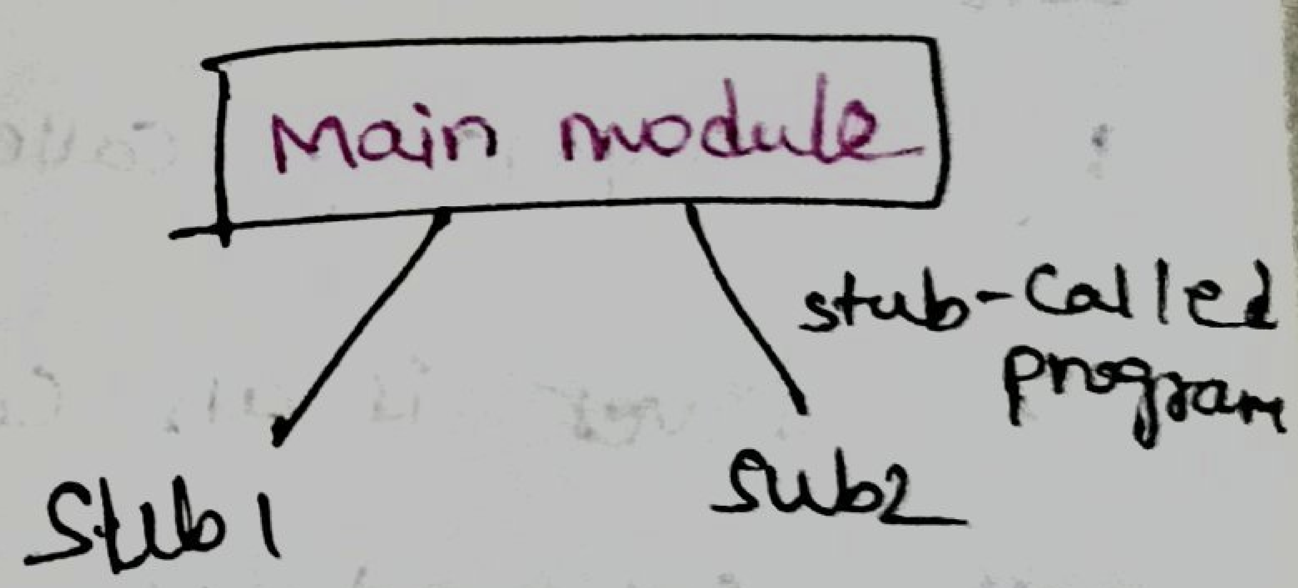
Mutation coverage:-

The developers mainly concentrate on the test coverage in this technique here the developers will introduce errors in the code and make sure the test cases written by the testers will identify the bugs introduce by the developers mainly here the team will concentrate on the test coverage.

Integration testing:-

Testing the connection b/w two modules are b/w integrated applications is called Integration testing. The Integration testing Performed in different ways.

1) Top down approach.
2) Bottomup approach.
3) Sandwich approach.
4) Big bang approach.



1) Top down Approach:-

In the top down approach the main modules are developed first followed by sub modules in few of the cases when submodules are not develop or construction the team will introduce a temporary program called stub

→ stub is an called Program because it been called by the main modules

3) Sandwich approach :-

In this approach we have both under construction sub modules & main modules.

→ The main module will be replaced by drivers which is an calling program and sub-modules are replaced by stub. which is an called Program.

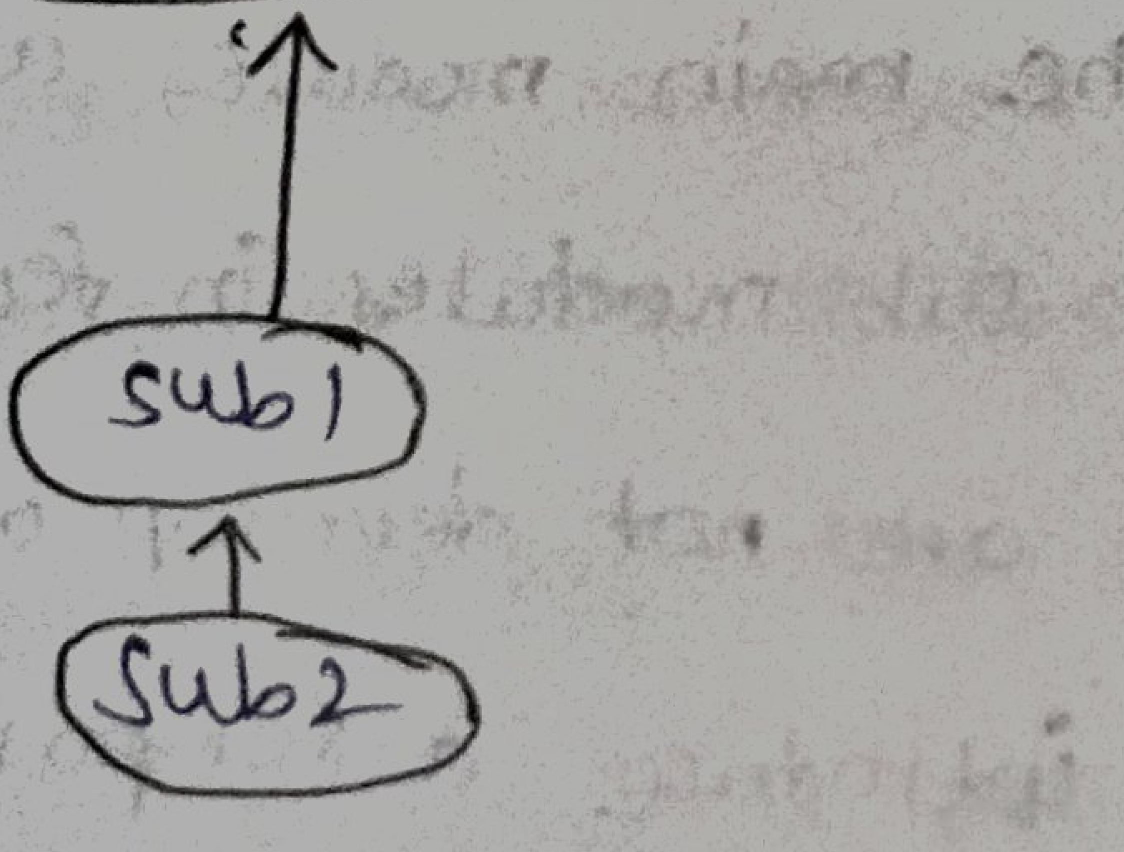
4) Bigbang approach :-

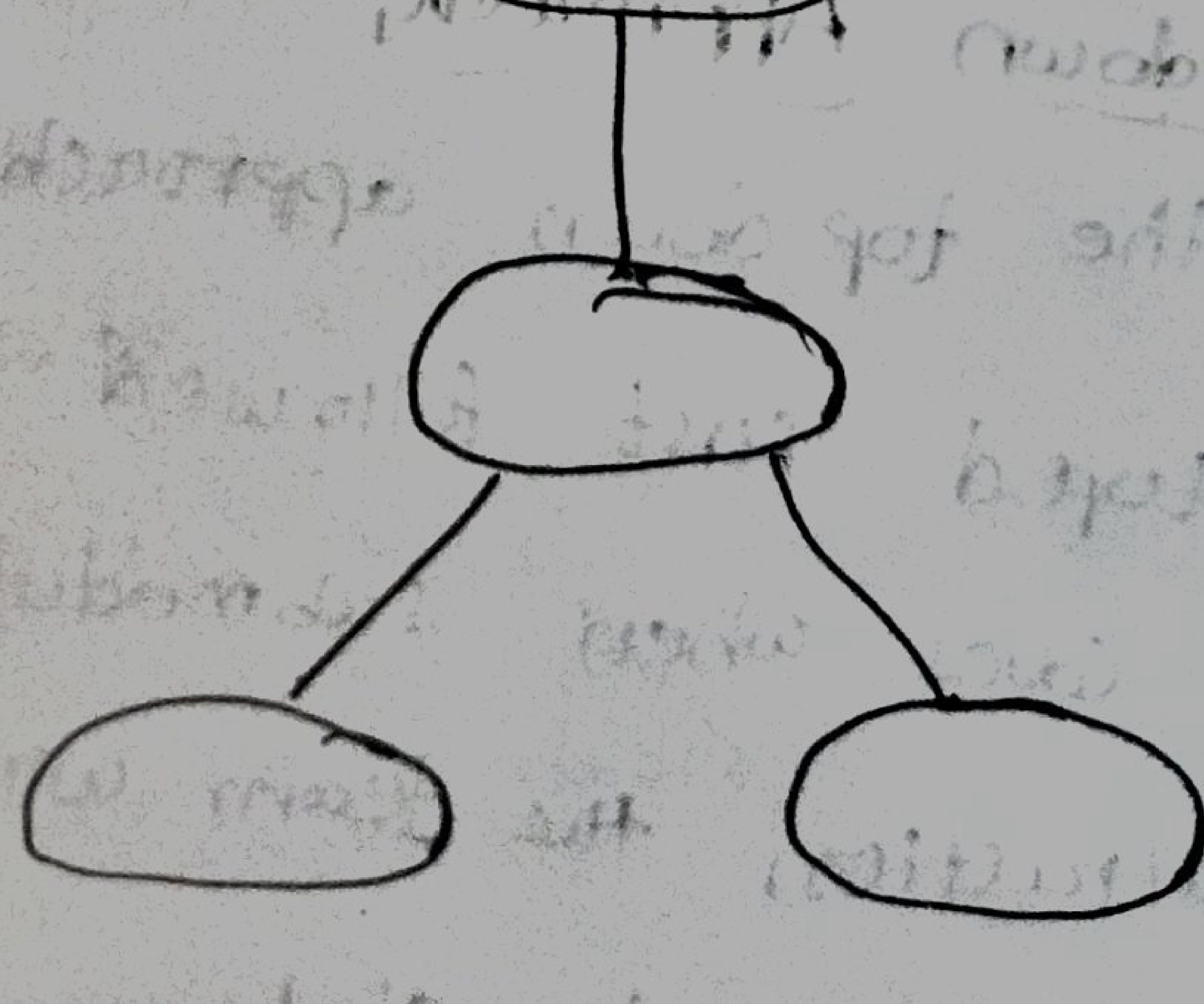
once the 100 % integration is done the team will release build to the test environment.

2) Bottom-up Approach :-

In this module the development will start submodules to main module when the main module is under construction or under development it will be replaced by a temporarly called driver.

The driver is an calling program as it is calling the submodules.





(~~Functional testing:-~~) Software testing:-

once the integration testing is done the developers will release build to the test environment.

→ The QA team need to verify the functionality of the application based on the requirements provided by the client.

→ The software testing is characterized into two types

(1) functional testing.

(2) Non-functional testing.

(1) Functional testing:-

Functional testing is mainly used to test the functionality of the application based on the requirements provided by the client.

→ The following are the different types of functional testing the team is going to perform once the user stories push to the test environment.

(i) GUI testing:- (Graphical user Interface)

In this testing the team will check the user interface such as elements interactable like text boxes, radio buttons, check boxes, links, images, alignments of the UI elements.

(ii) Input domain testing:-

In this testing the team will provide the random values of data to check how the system is behaving mainly in this testing the team will prepare different input data based on the requirements.

this testing is also called as 'fuzz' testing.

ex:- →Password should be min 8 characters &, max 16 characters,

→ first letter should be capital letter,

→ special characters like $,!,@,&.

→ should be alphanumeric 0 to 9

→ it should not your previous repeated five passwords.

→ there should not be spaces in the password.

test data →(test + data)

Login@123.

1) sirisha suresh@456 ---fail --- exceeds the max character.

2) sirisha rithu$221 --- fails -- fails the criteria(1st letter capital)

3) sirisha@rithu --- fail --- fails the criteria (no numerics)

4) Sirisha!124 --- pass

5) Sirisha rithu123 --- fail --- fails the criteria (spaces are included).

6) Sirisharithu12@5 --- pass --- max length.

7) SIRISHA@123 --- fail --- fails the criteria (all letters are caps)

8) 123456 --- fail

9) ChanduBudi12321 -- fail -- (spl not included & middle caps)

10) 123456789@ --- fail --- fail (no alphabets)

11) BuDi@& --- fail --- less than min,

12) Madhavi$$234 --- pass

13) special characters,

14) empty error.

15) Login #123

16) s@ha12ra!nya

(iii) Error handling:-

In this testing the team will Concentrate on errors, by providing an invalid data and checking whether proper error message is displaying or not.

→ The error messages will display in red Color text and shoud be validated against the user story

(iv) output domain (or) functional testing:-

In this testing the team will Perform different actions in the application based on the requirements provided by the client.

→ In this team mainly will focus whether the actual result met the Expected result,

(a) Database testing:-

In this testing the team will Provide data is the UI and the same data being Validated against the data base.

→ we have different databases such as SQl, noSQl, MongoDB, MySQl

(b) Smoke testing:-

→ Smoke testing is also called as build verification testing. This testing is mainly used to verify the health check of the build or stability of the build once new set of changes deployed to the test environment.

Mainly here the team will focus on the core functionality of the application

eg:- Flipkart

In the flipkart whenever new set of changes deployed the team will check whether user can able to login, perform such operation, select the product, add to the cart, place order, payment, check whether order placed successfully.

(c) Sanity testing:-

sanity is also an subset of regression. here, will check on an high level the newly deployed changes working or not.

→ In order to continue with further testing.

eg:- when update functionality is introduced as a part of sanity team will check a positive flow whether update is working before executing the other scenarios. (Happy path)

(d) Re-testing:-

retesting mainly used to verify the bugs

For eg:- one of the user story have 10 test cases when user execute this test cases, if they see any of the test cases got failed team will raise a bug and assign to the dev team and once it is fixed team will perform testing again to makesure the previous functionality is working fine.

(e) Regression testing:-

Testing the application whenever any changes are introduced here, the team will focus on existing functionality to make sure it is working as expected after adding the new functionalities to the application.

→ The Regression will be performed during below

1) whenever there is an Changes to the application
2) whenever functionality got removed.
3) Bug fixing
4) Environment changes.

Non-functional testing:-

checking the non-functional aspects of the application. can be conside) The following are the diff types of non-functional testing:-

(1) performance testing:-

checking the speed in process of an application is ~~turne~~ termed as Performance testing.

The below are the diff types of Performance testing.

(a) Load testing:-

The testing is performed based on the customer expected load with Customer expected environment to find load of the application.

(b) Stress testing:-

In stress testing the testing will be performed in customer expected environment and by increasing customer expected load to identify the peak load of the application

(c) spike testing:-

This testing is performed in customer expected environment but by sudden increasing the peak load of the application to identify the application crashing point.

(d) endurance testing:-

This is also called as soak testing.
The testing is performed with customer expected load and customer expected environment for longer duration in order to identify the data bridges or security bridges.

Security testing:-

This testing is also called as penetration testing where users will mainly concentrate on authorization, authentication, encryptions.

eg:-
→ everytime user should able to login with valid username & valid password into the application.
→ Always the password should be encrypted.
→ when user tries login into the application with the invalid credentials more than 3 times the account should be locked.

Compatability testing:-

Testing the application in different versions of browsers, os, platforms, devices etc:- to check or compare the functionalities

we have two diff types of compatability testings.

1) forward compatability testing,

2) Backward " "

→ In forward Compatability testing we are going to compare the functionality of the applications in the future versions.

→ In Backward compatability testing we are going to compare the functionality in older versions of the browsers, platforms, or etc:-

Multi languity:-

Testing the application with multiple languages,

There are two different types

(i) globalization

(2) Localization.

→ when the application is supporting multi languages then that application is considered as Globalize application,

→ when applications are supporting only local languags those applications are called as localize applications.

Recovery testing:-

Testing an application how it is going to recover if any crashes or any failures happen at application level.

Hardware Configuration testing:-

Testing the application with different hardware configuration such as printers, scanners etc:-

Usability testing:-

In the usability testing the team will mainly concentrate on the application how easy it is used and how easy it is understandable by the end-user to perform different actions in the application.

Acceptance testing:-

once the testing is performed by the QA team the same changes testing will be performed by BA team as BAT (Business Acceptance testing) and followed by UAT (user acceptance testing)

we have 3 diff types of acceptance testing:-

1) Alpha:-

when the sprint demo is provided by the testing team to the customer. then it is termed as alpha testing.

2) Beta:-

In Beta testing the sprint will be released to the test environment where customer will be going to perform the testing.

3) Gamma:-

In the gamma testing the sprint demo will be provided by the QA team & followed by the customer.

Release testing:-

This testing is performed by the release team during the production deployment.

Here, the team will concentrate on the newly set of changes that has been pushed to the production.

and check whether the functionalities are doing as per the expected behaviour.

Support and maintainance:-

once the release is done if there are any variations in the functionality or bugs or if user not able to understand the functionality. there is an dedicated support team which resolve this issue and test and deploy back to the production environment.